Project

The objective of this project is to investigate a physics problem that cannot be solved easily using analytical methods alone and therefore requires computational techniques.

Working in groups of 2–3 students, you should:

1. **Choose a physics problem** whose solution requires numerical computation, simulation, or data analysis. The problem may come from any area of physics/engineering, such as classical mechanics, electromagnetism, electronics, relativity, etc.

2. **Explain the physics problem** and clearly describe why computational methods are necessary. Discuss the limitations of analytical approaches and motivate the need for numerical calculations or simulations.

3. **Develop and implement an appropriate computational method** to solve the problem. This may involve numerical integration, root finding, differential equation solvers, Monte Carlo techniques, matrix methods, etc.

4. **Present and analyze the results.** Use figures, tables, and visualizations where appropriate. Discuss the physical interpretation of the results and evaluate the strengths and limitations of the computational method used.

5. **Explore at least one extension or parameter study** to demonstrate a deeper understanding of the problem and the computational techniques involved.

You should make a presentation that should include all the above points. The emphasis of the project is on understanding both the underlying physics and the role of computation in solving complex scientific/engineering problems.